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6. WASTE MANAGEMENT IMPLICATION

6.1 Introduction

6.1.1 This section identifies the types of wastes that are likely to be generated from the construction activities of the Project and evaluates the potential environmental impacts that may result from waste generated. It is unlikely for the operation of the Project to generate considerable amount of waste, adverse environmental impacts are thus not anticipated.

6.1.2 Mitigation measures and good site practices regarding waste handling, storage, collection and disposal are recommended in accordance with relevant waste legislation and guidelines. Handling and reuse of sediments are also discussed in this section.

6.2 Environmental Legislation, Standards and Guidelines

6.2.1 The criteria and guidelines for assessing waste management implications are outlined in Annexes 7 and 15 respectively in the Technical Memorandum on Environmental Impact Assessment (EIAO-TM).

6.2.2 The following legislation relates to the handling, treatment and disposal of wastes in the Hong Kong Special Administrative Region (SAR):

- Waste Disposal Ordinance (Cap. 354);
- Waste Disposal (Chemical Waste) (General) Regulation (Cap. 354C);
- Waste Disposal (Charges for Disposal of Construction Waste) Regulation (Cap. 354N);
- Dumping at Sea Ordinance (Cap. 466);
- Land (Miscellaneous Provisions) Ordinance (Cap. 28); and
- Public Health and Municipal Services Ordinance – Public Cleansing and Prevention Nuisances Regulation (Cap. 132BK).

Waste Disposal Ordinance (Cap. 354)

6.2.3 The *Waste Disposal Ordinance* (*WDO*) prohibits the unauthorised disposal of wastes. Construction waste is defined under Cap. 354N of the WDO as any substance, matter or thing that is generated and abandoned from construction works regardless of whether it has been processed or stockpiled before being abandoned, excluding sludge, screenings or any matter removed or generated from desludging, desilting or dredging works. Under WDO, waste can be disposed of only at designated waste disposal facilities licensed by the Environmental Protection Department (EPD).

Waste Disposal (Chemical Waste) (General) Regulation (Cap. 354C)

6.2.4 Issued under the WDO, the *Chemical Waste (General) Regulation* administers the possession, storage, collection, transport and disposal of chemical waste. EPD has also issued three statutory guidelines:-

- A Guide to the Chemical Waste Control Scheme (2016) to introduce and explain the legislative controls over the management of chemical waste in Hong Kong;

- A Guide to the Registration of Chemical Waste Producers (2024) to introduce the registration provisions of the Waste Disposal (Chemical Waste) (General) Regulation (the Regulation) and the procedure for identifying chemical waste generation; and

- Code of Practice on the Packaging, Labelling and Storage of Chemical Wastes (1992) to detail how the Contractor should comply with the regulations on chemical waste.

Waste Disposal (Charges for Disposal of Construction Waste) Regulation (Cap. 354N)

6.2.5 Under the *Waste Disposal (Charges for Disposal of Construction Waste) Regulation*, construction waste delivered to a landfill for disposal must not contain more than 50% by weight of inert material. Construction waste delivered to a sorting facility for disposal must contain more than 50% by weight of inert material, and construction waste delivered to a Public Fill Reception Facility (PFRF) for disposal must consist entirely of inert material.

Dumping at Sea Ordinance (Cap. 466)

6.2.6 The *Dumping at Sea Ordinance* (DASO) came into operation in April 1995 and empowers the Director of Environmental Protection (DEP) to control the disposal and incineration of substances and particles at sea for the protection of the marine environment. Under the Ordinance, a dumping permit from the DEP is required for the disposal of regulated substances within and outside the waters of Hong Kong. The permit contains terms and conditions which include the following specifications, but not limited to:-

- Type and quantity of substances permitted to be dumped;

- Location of the disposal grounds;

- Requirement of equipment for monitoring the disposal operations; and

- Environmental monitoring requirements.

6.2.7 Marine disposal of any dredged/excavated sediment is subject to control under the DASO. Section 4.2.1 of the Project Administration Handbook for Civil Engineering Works (PAH) stipulated sediment quality criteria or Chemical Exceedance Levels (CEL) for contaminants including metals, metalloid and organic pollutants. The PAH sets out the procedure for seeking approval to and the management framework for marine disposal of dredged/excavated sediment. This Practice Note outlines the requirements to be followed in assessing and classifying the sediment and explains the marine disposal arrangement for the classified material. The sediment quality criteria for the classification of sediment were referred as the Lower Chemical Exceedance Level (LCEL) and Upper Chemical Exceedance Level (UCEL). The LCEL and UCEL are presented in Appendix 4.28 of PAH. Subject to the results of the chemical screening, biological screening may be required to determine the disposal requirement of the sediment. There are three types of disposal options for dredged / excavated sediments stipulated in the Appendix 4.30 of PAH:-

- Type 1 – Open Sea Disposal or Open Sea Disposal in Dedicated Sites;

- Type 2 – Confined Marine Disposal; and

- Type 3 – Special Treatment / Disposal.

6.2.8 For Type 3 disposal, the Project Proponent shall be responsible for identifying and agreeing with the DEP the most appropriate treatment and/or disposal arrangement.

Land (Miscellaneous Provisions) Ordinance (Cap. 28)

6.2.9 The inert portion of Construction and Demolition (C&D) materials (including rocks, soil, broken concrete, building debris etc.) may be taken to PFRFs operated by the Civil Engineering and Development Department (CEDD). PFRFs usually form part of land reclamation schemes. The Land (Miscellaneous Provisions) Ordinance requires that individuals or companies who deliver public fill to the PFRFs are required to obtain Dumping Licences. The licences are issued by CEDD under delegated authority from the Director of Lands.

Public Health and Municipal Services Ordinance – Public Cleansing and Prevention of Nuisances Regulation (Cap. 132BK)

6.2.10 The Public Cleansing and Prevention of Nuisances Regulation provides control on illegal dumping of wastes on unauthorised / unlicensed sites. The illegal dumping of wastes can lead to a fine and/or imprisonment.

Other Relevant Environmental Guidelines

6.2.11 Other environmental relevant circulars/guidelines applicable to waste management practices for this Project include:-

- Works Branch Technical Circular (WBTC) No. 2/93 Public Dumps;
- WBTC No. 2/93B Public Filling Facilities;
- WBTC No. 12/2000 Fill Management;
- Environment, Transport and Works Bureau Technical Circular (Works) (ETWB TCW) No. 19/2005 Environmental Management on Construction Sites;
- Development Bureau Technical Circular (Works) (DevB TCW) No. 06/2010 Trip-ticket System for Disposal of Construction and Demolition Material;
- DevB TCW No. 08/2010 Enhanced Specification for Site Cleanliness and Tidiness;
- DevB TCW No. 09/2011 Enhanced Control measures for Management of Public Fill;
- DevB TCW No. 4/2020 Tree Preservation;
- CEDD Technical Circular (TC) No. 11/2019 “Management of Construction and Demolition Materials”;
- Hong Kong Planning Standards and Guidelines, Chapter 9 - Environment;
- Code of Practice on the Packaging, Labelling and Storage of Chemical Wastes (EPD);
- A Guide to the Chemical Waste Control Scheme (EPD);

- PAH for Civil Engineering Works, Section 4.1.3 of Chapter 4 "Construction and Demolition Materials" (subsumed from WBTC Nos. 25/99, 25/99A, 25/99C "Incorporation of Information on Construction and Demolition Material Management in Public Works Subcommittee Papers" and ETWB TCW No. 33/2002 "Management of Construction & Demolition Materials Including Rock)";

- PAH for Civil Engineering Works, Section 4.2.1 of Chapter 4 "Management of Dredged/Excavated Sediment" (subsumed from ETWB TCW No. 34/2002 "Management of Dredged/Excavated Sediment");

- Guidance Note No. 1/2024" under the Dumping at Sea Ordinance.

- Relevant guidelines on handling of yard waste on EPD's website (https://www.epd.gov.hk/epd/english/environmentinhk/waste/manage_facility/ypark.html) and Y·Park's website (https://www.ypark.hk/zh-hant/); and

- The Greening, Landscape and Tree Management Section of the Development Bureau "Guidelines on Yard Waste Reduction and Treatment".

6.2.12 The WBTC No. 2/93 'Public Dumps' documents the current policy related to the disposal of C&D materials. C&D materials that are wholly inert, i.e. public fill, should be taken to PFRFs for reuse instead of being disposed of at landfill.

6.2.13 Under DevB TCW No. 6/2010 'Trip Ticket System for Disposal of Construction and Demolition Materials', for all contracts that are expected to generate inert C&D materials requiring disposal from site, the project office shall write to the Public Fill Committee (PFC) through Secretary of the PFC to request a designated disposal ground for incorporation into the tender documents. For contracts where the estimated amount of non-inert C&D materials requiring disposal at landfill facilities equals to or exceeds 50 m^3, the project office shall seek confirmation from the DEP in terms of the availability of landfill facilities for disposal of such materials and the DEP will designate landfill facilities, if available, for the contracts. For contracts where the estimated amount of non-inert C&D materials to be generated from the contract is less than 50 m^3, the project office is not required to apply to DEP for designated landfill facilities but it should still specify in the tender documents of the appropriate landfill facilities for disposal.

6.2.14 Further measures have been introduced under Paragraph 4.1.3 of Chapter 4 of the PAH for Civil Engineering Works to enhance the management of C&D materials, and to minimise its generation at source. The enhancement measures include:-

- Drawing up a Construction and Demolition Materials Management Plan (C&DMMP) at the feasibility study or preliminary design stage which generates more than 50,000m^3 of C&D materials including rock or which requires imported fill in excess of 50,000m^3 in order an early design stage to minimise C&D materials generation and to encourage proper management of such materials;

- Drawing up a C&DMMP where more than 300,000 m^3 of surplus inert C&D materials are generated and submission of the plan to the Public Fill Committee (PFC) for in-principle approval prior to the commencement of detailed design;

- Vetting of the C&DMMP prior to upgrading of the project to Category A in the Public Works Programme; and

- Providing the contractor with information from the C&DMMP in order to facilitate the preparation of Waste Management Plan (WMP) and to minimise C&D materials generation during construction.

6.2.15 The *ETWB TCW No. 19/2005 Environmental Management on Construction Sites* introduces additional measures to enhance waste management on construction sites. The circular sets out the policies and procedures that require the contractors to prepare and implement an Environmental Management Plan (EMP), which includes the WMP to encourage on-site sorting of C&D materials and to minimise generation of C&D materials during the course of construction.

6.3 Description of the Environment

6.3.1 The Project encompasses the Loop and Ma Tso Lung (MTL) areas. The western section is located near the southeastern edge of the Loop and follows the Shenzhen Meander, traversing hilly terrain that includes Horn Hill (Ngau Kok Shan). On the eastern side, the Project covers the Ma Tso Lung area, which is adjacent to Shun Yee Shan Tsuen and extends to the east of Ma Tso Lung San Tsuen.

6.4 Identification of Waste Management Implications

Construction Phase

6.4.1 The construction of the Project will tentatively commence in Q1/Q2 2027, with anticipated completion of works in Q4 2031. The construction activities to be carried out for the Project include site clearance, slope work, excavation, piling works, backfilling, foundation works, road works and pavement etc. The types of waste associated with these activities include:-

- C&D materials ;
- Excavated sediments
- Chemical wastes; and
- General refuse.

Operation Phase

6.4.2 During the operation phase, it is anticipated that the volume of waste generated would not be significant however waste generated may include:-

- Chemical waste; and
- General Refuse

6.5 Assessment Methodologies

6.5.1 The assessment of the potential waste management implications during the construction phase of the Project has been conducted in accordance with Annexes 7 and 15 of the EIAO-TM, including the following tasks:-

- Estimation of the types and quantities of the wastes generated;
- Evaluation of opportunities for waste reduction and re-use;

- Identification of disposal options for each type of waste;
- Assessment of potential environmental impacts arising from waste management with respect of potential hazards, air and odour emissions, noise, wastewater discharge, and public transport; and
- Assessment of the impacts caused by handling, collection, transportation and re-use /disposal of wastes.

6.5.2 Prior to considering the disposal options for various types of waste, opportunities for reducing waste generation, on-site or off-site reuse and recycling have been evaluated. Measures which can be taken in the construction phases for maximising waste reduction have been separately considered. Practices to promote segregation of waste materials are additionally considered for advancing the waste management efficiency.

6.5.3 After considering all the opportunities for reducing waste generation and maximising reuse, the types and quantities of the waste required to be disposed of have been estimated and the disposal options for each type of waste have been described. The disposal method recommended for each type of waste has taken into account the result of the assessment. The impacts caused by handling (including stockpiling, labelling, packaging and storage), collection and reuse / disposal of waste have been addressed and appropriate mitigation measures have been proposed.

6.6 Prediction and Evaluation of Waste Implications

Construction Phase

Construction and demolition (C&D) materials

6.6.1 C&D materials would be generated from construction works of the Project such as site clearance, excavation/foundation/piling works, slope works and other construction activities. The C&D materials would comprise both inert C&D materials (i.e. public fill including soil, rock, concrete & etc.) and non-inert C&D materials (i.e. C&D waste including vegetation, timber, paper, plastic etc.). During general layout design, devising construction method described in **Section 2** and planning of construction programme, considerations have been given to minimise the generation of inert C&D materials and maximise the use of generated inert C&D materials for other construction works. The portion of the inert C&D materials generated from the excavation works will be reused on-site as backfilling materials in underpass and slope works. Considerations include optimisation of structural design by use of modular and prefabricated structural elements to minimise over-excavation and unnecessary concrete use. The viaduct section of ECR at the north-western and north-eastern side of Shun Yee San Tsuen will be constructed using precast concrete; to minimise the quantities of C&D waste / materials to be generated on-site.

6.6.2 Based on the latest layout in **60722948/A10/Figures 2.2 – 2.5**, the estimated cut and fill volume for each works section is provided in **Table 6.1**.

Table 6.1 Estimated Cut and Fill Volumes by Works Section

Works Section	**Cut Volume [1,3] (m^3)**	**Fill Volume [2,3] (m^3)**
At-grade road	0	0
Underpass	274,300	105,700

Works Section	**Cut Volume [1,3] (m^3)**	**Fill Volume [2,3] (m^3)**
Depressed Road	65,000	0
Viaduct [4]	600	0
Slope formation works [4]	68,000	10,700
Total	**407,900**	**116,400**

Notes: 1. Cut volume include sediment (as identified in **Table 6.5**) and inert C&D materials (as identified as "Generated during construction phase" in **Table 6.3**)
2. Fill volume include reuse of treated sediment (as identified in **Table 6.5**)and inert C&D materials (as identified as "Reuse on Site " in **Table 6.3**)
3. Totals may not equal the sum of individual due to rounding error.
4. Cut volumes for viaduct and slope formation works only include inert C&D materials and do not include sediment.

6.6.3 The estimated cut and fill volumes for the Project by year are shown in **Table 6.2**. Major cut volume is expected from underpass, depressed road and slope formation works. The programme was planned with consideration to prioritise the reuse of inert C&D materials and resulted no major programme mismatch on cut and fill. The Contractor should review the programme during early construction stage to maximise the quantity of on-site reuse of surplus fill materials. Based on the estimation, no imported fill is required for the Project.

Table 6.2 Estimated Cut and Fill Volumes for the Site Clearance and Site Formation Works by Year

Year	**Cut Volume [1,3] (m^3)**	**Fill Volume [2,3] (m^3)**
2027	65,800	4,300
2028	79,700	65,200
2029	124,200	38,000
2030	57,200	8,900
2031	81,100	0
Total	**407,900**	**116,400**

Notes: 1. Cut volume include sediment (as identified in **Table 6.5**) and inert C&D materials (as identified "Generated during construction phase" in **Table 6.3**)
2. Fill volume include reuse of treated sediment (as identified in **Table 6.5**)and inert C&D materials (as identified "Reuse on Site " in **Table 6.3**)
3. Totals may not equal the sum of individual due to rounding error.

6.6.4 The total quantity of C&D materials generated during the construction phase of the Project (anticipated to commence in 2027 and be completed by 2031) is estimated to be 313,100 m^3 with 5,900 m^3 of non-inert materials and 307,200 m^3 of inert materials. These quantities are initial estimates only and will be reviewed and updated as the design progresses. A C&DMMP would be prepared in accordance with Section 4.1.3 of the PAH and submitted to PFC for approval in the detail design stage.

6.6.5 The inert C&D materials should be reused on-site as far practicable subject to the review at detailed design stage before being exported off site or delivered to PFRF (e.g. Tuen Mun Area 38 Fill Bank) as designated by PFC for beneficial use in other projects with potential fill demand. The amount of inert materials generated during the

construction phase is estimated to be around 307,200 m^3 of which approximately 15,700 m^3 is estimated to be reused on-site as backfill material. With reference to **Section 6.6.22**, another 100,700m^3 of treated sediment is estimated to be reused on-site as backfill material. The underpass / depressed road section of the project will be constructed by cut-and-cover method. Materials excavated during the construction of the underpass / depressed road will as far as practical be reused as backfilling materials after completion of structure, however a surfeit of inert C&D materials generated may require transport for beneficial off-site use. Compaction of the backfilling materials shall follow the requirement stated in the latest edition of General Specification for Civil Engineering Works.

6.6.6 For non-inert C&D materials, upon segregation of non-recyclables and exhaustion of recycling efforts, these materials would be disposed of to landfill as designated by EPD. Yard Waste will be segregated from other non-inert C&D materials with tree trunks and branches sent to Y·PARK for recycling. The amount of non-inert materials generated during the construction phase is estimated to be around 5,900 m^3. The non-inert C&D materials should be recycled or reused on-site as much as possible before disposing at the NENT or WENT Landfill. It is the Contractor's responsibility to separate the inert and non-inert C&D materials on site. A stockpiling area should be assigned to facilitate the sorting of inert and non-inert C&D materials and store the inert C&D materials for subsequent backfilling. Limited spatial area is available within the Project boundary and subject to actual on-site conditions, land required for stockpile area would be minimised as far as possible. Tentative stockpile area is shown in **60722948/A10/FIGURE 6.2**. The exact location/size and availability of the stockpile area is subject to contractor progamme and actual site condition.

6.6.7 Where yard waste generation is unavoidable, sorting of yard waste for recycling and reuse on-site should always be prioritised. Yard waste shall be separated from C&D materials to facilitate recycling such as delivering them to Y·Park so as to minimise the quantity of waste to be disposed of at landfill site. Where appropriate, the Contractor should be responsible to cut and shred the yard waste in order to meet the collection requirement of the recycling outlet for processing. Disposal of yard waste directly at landfills should only be regarded as the last resort when no alternatives are available.

6.6.8 A summary of the estimated quantities of C&D materials generated from the site clearance, excavation/foundation works, slope works, piling works and other construction activities together with the estimated volumes of inert C&D material to be delivered to public fill reception facilities designated by CEDD and estimated volume of non-inert C&D material to be disposed of landfill during construction phase are presented in **Table 6.3**.

Table 6.3 Estimated Quantities of C&D Materials during Construction Phase

	Estimated Amount of inert C&D Materials [1,2] (m^3)	**Estimated Amount of non-inert C&D material [1,3] (m^3)**
Generated during construction phase	307,200	5,900
Reuse on Site	15,700	-
Deliver to Public Fill Reception Facilities	291,500	-

	Estimated Amount of inert C&D Materials [1,2] (m^3)	**Estimated Amount of non-inert C&D material [1,3] (m^3)**
Dispose of at Landfills	-	5,900

Notes: 1. Estimated C&D materials does not include sediment.
2. Inert C&D materials includes soil, rock, concrete
3. Non-inert C&D materials includes vegetation, timber, papers & plastics

6.6.9 The non-inert C&D materials would be disposed of at the North East New Territories (NENT) Landfill or West New Territories (WENT) Landfill while the surplus inert C&D materials (general fill) are proposed to be transported to Tuen Mun Area 38 Fill Bank or other construction sites with potential fill demand via the tentative transportation routing presented in **Table 6.7**. Disposal site of inert C&D materials is subject to the designation by the Public Fill Committee according to DEVB TC(W) No. 6/2010. The Contractor should liaise with CEDD on the arrangement of and obtain all necessary permits on disposal at the fill bank. The average number of trip and the number of trip per day in peak generation quarter is provided in **Table 6.4**. With the implementation of measures recommended in **Section 6.7**, no adverse environmental impacts (including potential hazard, air and odour emissions, noise and wastewater discharge and ecology) and public transport impact caused by handling, collection, transportation and reuse/disposal of C&D materials are expected.

Table 6.4 Number of Trips for Different Waste Types Anticipated during Construction Phase

Outlet	**Type of waste**	**Estimated Overall Quantity to be delivered/ disposed (m^3)**	**Average No. of Trips per day [3,4]**	**Estimated Peak Quarterly Quantity to be delivered / disposed [1,2] (m^3)**	**Maximum No. of Trips per day in Peak Generation Quarter [3]**
PFRF	Inert C&D materials	291,500	25	18100 (2029 Q2)	31
Landfill	Non-Inert C&D materials	5,900	1 [5]	660 (2027 Q1 – Q4)	2 [5]

Notes: 1. The "Peak Quarterly Quantity to be delivered / disposed" during the entire construction period is tentative at this stage, the actual "Peak Quarterly Quantity to be delivered / disposed" will be subject to site condition and contractor arrangement.
2. Assume 26 working days per month / Assume 78 working days per quarter (three months)
3. Dump truck is assumed to be heavy vehicles with permit gross vehicle weight at 30 tonnes. The capacity is assumed to be 15 tonnes / 7.5 m^3.
4. Average No. of trip is based on 60 months construction
5. General refuse (estimated daily rate of 65 kg of general refuse) is assumed to be absorbed in the trip to landfill

Excavated Sediment

6.6.10 Sediment is anticipated to be excavated during the construction of underpass / depressed road section with reference to **60722948/A10/FIGURE 2.2** and **2.3**. Sediment is not anticipated at the depressed road section near LMC Operational Base.

The potential location of sediment is provided in **60722948/A10/FIGURE 6.1** and potential sediment is anticipated from the following:

- Pond sediment / Swamp Deposits;
- Marine deposits; and
- Estuarine deposits.

6.6.11 The total volume of sediment generated is estimated to be approximately 100,700 m^3. A breakdown on sediment volume estimation is provided in **Table 6.5**. The extent of intertidal deposits references the 1:20,000 Geological Map Sheet 2. Potential pond sediments are also identified in the vicinity. The extent of various sediment types are provided in **60722948/A10/FIGURE 6.1**.

Table 6.5 Estimated Volume of Sediment Generation

Location	Sediment type	Estimated Volume of Sediment Generation (m^3)
Underpass section to the east of LMC Meander [1]	Pond Sediment	9,100
	Marine Deposits	33,600
	Estuarine deposits	6,300
Underpass section to the west of LMC meander and at LMC meander [2]	Swamp Deposits [4]	39,300
Depressed road section at the Loop [3]	Estuarine deposits [4]	12,400
	Total	100,700

Notes:
1. The estimated sediment volume for underpass section to the east of LMC Meander was approximated based on the average sediment thickness from drillholes (sediment thickness at ENV-BH1 and S8F/DH9 are 12.9m and 0.0m respectively, see **Appendix 6.3**. Average sediment thickness = ½ (12.9m + 0.0m)) and the extent of intertidal deposits overlapping with the underpass section, approximately 7,590 m^2, as illustrated in **60722948/A10/FIGURE 6.1**.

2. The estimated sediment volume for underpass section to the west of LMC meander and at LMC meander was approximated based on a reference drillhole (A-DH14 with sediment thickness of 10.2m, see **Appendix 6.3**) and the area of the underpass section, approximately 3850 m^2, as illustrated in **60722948/A10/FIGURE 6.1**.

3. The estimated sediment volume for depressed road section at the Loop was approximated based on the average sediment thickness from drillholes a reference drillhole (ADH3(P) with sediment thickness of 2.9m, see **Appendix 6.3**) and the area of depressed road section, approximately 4280 m^2, as illustrated in **60722948/A10/FIGURE 6.1**.

4. This may contain other sediment including pond sediment, swamp deposits and estuarine deposits as the Loop was filled during previous river training.

6.6.12 Previous ground investigation record was reviewed. One additional drillhole (ENV-BH1) was constructed in August 2025 to collect sediment quality information. Drillhole record has been provided in **Appendix 6.3**. The location of drillhole are provided in **60722948/A10/FIGURE 6.1**.

6.6.13 The drillhole ENV-BH1 was located approximately 12m from the edge of Project boundary. Due to existing site constraints including private ownership of most fish pond areas, presence of soft ground and elevated water level, it was not feasible to conduct

sampling within fish ponds. Given these limitations, the ENV-BH1 is considered representative for the Project site. The vertical profile of sediment from ENV-BH1 is as follows:

- Pond sediment: +0.59 mPD to +2.99 mPD
- Marine deposits: -8.26 mPD to +0.59 mPD
- Estuarine deposits: -9.91 mPD to -8.26 mPD

6.6.14 Drillholes S8F/DH9, B-DH9, SBF/DH8 and trial pit SBF/TP38 were located along Border Road. Fill, alluvium/colluvium and meta-sandstone were encountered in these drillholes and trial pits. No sediment was encountered. Consider these boreholes with ENV-BH1, it is expected that sediment along the underpass section to the east of LMC Meander will be diminishing from ENV-BH1 towards the border road.

6.6.15 With reference to the Development of Lok Ma Chau Loop EIA Report (EIA Register No. AEIAR-176/2013), seven sediment grab samples (GR5 – GR11) were collected from the old meander of Shenzhen River. The locations and the chemical screening results for sediment samples GR5 – GR11 are provided in **Appendix 6.1** with the location of sediment samples GR5 – GR11 provided in **60722948/A10/FIGURE 6.1**. All samples (GR5 – GR11) are classified as Category H with Nickel and Zinc exceeding the Upper Chemical Exceedance Level (UCEL).

6.6.16 Testing results have been reviewed in accordance with the Appendix 4.28 of PAH, sample classifications and disposal options as summarised in **Table 6.6**. Detailed sediment categorisation is provided in **Appendix 6.2**.

Table 6.6 Classification and Disposal Options for Sediment Samples

Sediment Sample [1]	Sediment type	Category [2]	Disposal Option
GR5 (Grab)	Sediment from Meander	H	Type 2 – Confined Marine Disposal
GR6 (Grab)	Sediment from Meander	H	Type 2 – Confined Marine Disposal
GR7 (Grab)	Sediment from Meander	H	Type 2 – Confined Marine Disposal
GR8 (Grab)	Sediment from Meander	H	Type 2 – Confined Marine Disposal
GR9 (Grab)	Sediment from Meander	H	Type 2 – Confined Marine Disposal
GR10 (Grab)	Sediment from Meander	H	Type 2 – Confined Marine Disposal
GR11 (Grab)	Sediment from Meander	H	Type 2 – Confined Marine Disposal

Notes: 1. All sample GR5 – GR11 are grab samples.
2. No biological screening test was conducted.

6.6.17 With reference to the Development of Lok Ma Chau Loop EIA Report (EIA Register No. AEIAR-176/2013), the Loop (i.e. former agricultural land and fish ponds in 1980s and early 1990s) was used as a dumping ground for mud extracted from Shenzhen River

Training Works Stages 1 and 2. Various sediment was encountered at various level in the Loop. A few drillhole in the Loop was referred as below:

- A-DH14 with swamp sediment: -7.79 mPD to +2.41 mPD
- ADH3(P) with estuarine deposits: -0.71 mPD to +2.19 mPD
- ADH5(M) with estuarine deposits: -5.08 mPD to -2.88 mPD
- ADH8(V) with pond deposits: +2.01 mPD to +4.01 mPD

6.6.18 It is expected that sediment along underpass section to the west of LMC meander and at LMC meander and the depressed road section at the Loop is disturbed and of various sediment types (pond sediment, swamp deposits and estuarine deposits).

6.6.19 With respect to sediment quality for pond sediments, metal-based algaecides including copper-based algaecides are not commonly used for freshwater fish farming in Hong Kong due to the high cost and toxicity to fish. Calcium oxide, which is cheaper, non-toxic to fish and a more readily available chemical, is more commonly used to control algae and suspended solids within fish ponds. It is therefore considered unlikely that the pond sediment will be contaminated.

6.6.20 For estuarine deposits and marine deposits forming existing land outside the old meander of Shenzhen River, no major development has been identified since 1973. As the area has remained rural since 1973, both estuarine and marine deposits outside the old meander of Shenzhen River are not likely to be contaminated.

6.6.21 With no contamination source identified and as the area has been fish ponds since 1980s, it is considered that, apart from sediment in the old meander of Shenzhen River, various sediment types in the vicinity are not likely to be contaminated.

6.6.22 Sediment excavated during the project is proposed to be re-used onsite as far as practicable after treatment. Excavated sediment would be treated by cement stabilisation / solidification (S/S) and fully reused as backfilling materials within the Project site. Treated sediment will be preferentially reused over general inert C&D materials for backfilling purposes in the Project. In case reuse of sediment is not practical, marine disposal of the sediment will be required according to Section 4.2.1 of the PAH. Should marine disposal of sediment by unavoidable at later stage upon exhaustion of reuse options, separate submissions (e.g. Sediment Sampling and Testing Plan (SSTP) and Sediment Quality Report (SQR)) shall be submitted to EPD's Marine Dumping Control Section / Territorial Control Office if application of dumping permit under DASO is required. The rationale for sediment removal/disposal will also need to be submitted to the Marine Fill Committee (MFC)/CEDD for agreement in accordance with the PAH.

6.6.23 The transportation routing and frequency for marine disposal to designated disposal outlets shall be agreed with EPD/ MFC CEDD during application of DASO dumping permit prior to marine disposal. Detail on transportation vessel will be also submitted for approval during permit application.

Chemical Waste

6.6.24 The maintenance and servicing of construction plant and vehicles during construction phase of the Project (anticipated to commence in 2027 and be completed by 2031) may generate a negligibly small amount of chemical waste such as:

- Used paint, engine oils, hydraulic fluids and waste fuel from plant operation;
- Spent mineral oils/ cleaning fluids from plant maintenance; and
- Spent solvents/ solutions from equipment maintenance.

6.6.25 It is difficult to quantify the amount of chemical waste that would arise from the construction activities as it would be highly dependent on the Contractor's on-site maintenance activities and the quantity of plant and equipment utilised. It is anticipated that the quantity of chemical waste, such as lubrication oil and solvent produced from plant maintenance, would be small and in the order of a few hundred litres per month. The amount of chemical waste to be generated would be quantified in the WMP to be prepared by the Contractors. The disposal of chemical waste by licensed chemical waste collector is expected to require maximum 1 trucks per month for transporting the chemical waste. With the implementation of measures recommended in **Section 6.7**, no adverse environmental impacts and public transport impact associated with handling, collection, transportation and disposal of chemical waste is expected.

6.6.26 Materials classified as chemical waste shall require special handling and storage arrangements by the Contractor. All chemical waste shall be collected by a licensed chemical waste collector and be disposed at a licensed chemical waste treatment and disposal facility such as Chemical Waste Treatment Centre (CWTC) at Tsing Yi via the tentative transportation routing presented in **Table 6.7**. Unused chemical or those with remaining functional capacity would be reused and recycled on site or by licensed companies whenever possible. Mitigation and control requirements for chemical wastes are detailed in **Section 6.7**. Provided that the handling, storage and disposal of chemical wastes are in accordance with these requirements and that stipulated in the *Waste Disposal (Chemical Waste) (General) Regulations* and the *Code of Practice on Packaging, Labelling and Storage of Chemical Wastes* published by EPD, adverse environmental impacts (including potential hazard, air and odour emissions, noise, wastewater discharge and ecology) and public transport impact will be negligible.

General Refuse

6.6.27 General refuse comprising general site waste, such as packaging and container waste from materials and equipment used on the site, food scraps, waste paper, empty containers etc. would be generated by workforce during the construction phase of the Project (anticipated to commence in 2027 and be completed by 2031). Improper collection and removal of general refuse would give rise to hygiene problems and adverse environmental impacts to local residents, e.g. odour impacts, and contamination of the nearby watercourses.

6.6.28 The maximum number of construction workers to be working on site at any one time was estimated to be about 100. Based on a generation rate of 0.65kg per workers per day, approximately 65 kg of general refuse would be generated daily during the construction period. It is expected that the general refuse from construction sites would be disposed of at NENT or WENT landfill via the tentative transportation routing presented in **Table 6.7**. Given the small quantity of general refuse to be disposed of at landfill, no adverse impact on the operation of landfill would be anticipated. The disposal of general refuse is expected to require maximum 1 truck per day for transporting the general refuse. With the implementation of measures recommended **Section 6.7**, no adverse environmental impacts and public transport impact associated with handling, collection, transportation and disposal of general refuse is expected.

6.6.29 In order to minimise the final disposal quantities of general refuse, provision of sufficient number of recycling bins for the collection of different types of recyclable waste (including food waste, paper, aluminium cans, plastic and glass bottles) and sufficient number of general refuse bins for the collection of non-recyclable waste is recommended. The Contractor should implement an education programme for worker relating to avoiding, reducing, reusing and recycling of general refuse.

6.6.30 The general refuse will be collected on-site on a regular basis, separately from C&D materials by an appropriate waste collector to be employed by the Contractor. Prior to disposal off-site, such refuse will be temporarily put in suitably covered storage areas / bins where they will have to be regularly cleaned and maintained to avoid attracting vermin and pests. With proper on-site handling and storage as well as regular disposal of the wastes, no unacceptable environmental impact would be anticipated. Recommendations of mitigation measures for managing general refuse are presented in **Section 6.7**.

Transportation Arrangement for Waste Disposal during Construction Phase

6.6.31 Land transport should be used to deliver and dispose of the waste generated from the Project area to the designated disposal outlets. Referring to **Section 6.6.9**, It is estimated a daily average of 25 trip of dump truck for inert C&D to PFRF and 1 trip of dump truck for non-inert C&D materials to landfill. The tentative transportation routings for the disposal of various types of wastes generated during the construction phase of the Project are presented in **Table 6.7**. The transportation routings may change subject to the actual traffic conditions of the roads. Nevertheless, with the implementation of appropriate mitigation measures (e.g. using water-tight containers and covered trucks), no adverse environmental impacts are expected due to the transportation of waste.

6.6.32 No barging points or conveyor systems would be established in the Project area. The inert C&D materials will be reused on-site as much as practicable. Any surplus inert C&D materials would be stockpiled or transported out of construction work area. As presented in **Section 6.6.6**, subject to actual on-site condition, limited spatial area is available within the Project boundary, land required for stockpile area would be minimised as far as possible. Tentative stockpile area is shown in **60722948/A10/FIGURE 6.2**. Details will be reviewed in design stage, in case stockpiling areas are found required, they would be implemented with mitigation measures such as covering the materials during heavy rainfall.

Table 6.7 Tentative Transportation Routings to Various Outlets During Construction Phase

Outlet	Type of Waste	Tentative Transportation Routing
Tuen Mun Area 38 Fill Bank [1]	Surplus Inert C&D Materials	Via Lok Ma Chau Road, Castle Peak Road (San Tin), San Tin Highway, Yuen Long Highway, Tuen Mun Road, Wong Chu Road, Lung Fu Road, Lung Mun Road
CWTC [2]	Chemical Waste	Via Lok Ma Chau Road, Castle Peak Road (Chau Tau), Castle Peak Road (San Tin), San Tin Highway, Tsing Long Highway, Tsing Sha Highway, Tsing Yi Road

Outlet	Type of Waste	Tentative Transportation Routing
NENT Landfill [3] or WENT Landfill [3]	Non-inert C&D Materials and General Refuse	Via Lok Ma Chau Road, Castle Peak Road (San Tin), Fanling Highway, Heung Yuen Wai Highway, Wo Keng Shan Road Or Via Lok Ma Chau Road, Castle Peak Road (San Tin), San Tin Highway, Yuen Long Highway, Tuen Mun Road, Wong Chu Road, Lung Fu Road, Lung Mun Road, Lung Kwu Tan Road, Nim Wan Road
Y-park	Yard waste	Via Lok Ma Chau Road, Castle Peak Road (San Tin), San Tin Highway, Yuen Long Highway, Tuen Mun Road, Wong Chu Road, Lung Fu Road, Lung Mun Road, Lung Kwu Tan Road

Notes: 1. The exact destination of inert C&D materials is subject to the designation by the Public Fill Committee according to DEVB TC(W) No.6/2010
2. Chemical waste can be disposed of at CWTC or other licensed chemical waste treatment / disposal facilities.
3. The exact destination of non-inert C&D materials is subject to the designation by EPD

Operation Phase

6.6.33 During the operation phase, the volume of waste generated is not anticipated to be significant but include:

- Chemical waste; and
- General Refuse

Chemical Waste

6.6.34 Chemical wastes such as paints, lubricants and used batteries may be generated during maintenance activities. This waste may pose environmental, health and safety hazards. Measures as stipulated in the Waste Disposal (Chemical Waste) (general) Regulation and the Code of Practice on the Packaging, Labelling and Storage of Chemical Wastes would be strictly followed for the handling and disposal of chemical waste.

6.6.35 Should any chemical waste be generated, the operator should register with EPD as a chemical waste producer. The chemical waste disposal facilities include the Chemical Waste Treatment Centre (CWTC) at Tsing Yi and other appropriately licensed facilities. This chemical waste should be stored in drum-type containers and collected by licensed chemical waste collectors. The quantities of chemical waste generated from the operation phase is considered as minimal, a few cubic metres per month is anticipated and approximate 1 truck per month for transport disposal envisaged. With proper storage, handling and disposal of this waste, no adverse environmental impact is anticipated.

General Refuse

6.6.36 During the operation phase, general refuse would be generated during maintenance works. The waste includes food waste, paper, wood, plastic, packaging, etc. The amount of general refuse will be insignificant and will be removed during maintenance works. Plastics, papers and other recyclable wastes should be separated from general refuse and recycled as far as possible. The remaining refuse would be collected by contracted collectors. Given this small number of vehicle trips, no adverse impacts will be envisaged.

6.6.37 The anticipated potential environmental impacts arisen from the handling, storage and disposal of waste in operation phase would be insignificant provided that the mitigation measures stated in **Section 6.7** are strictly followed.

6.7 Mitigation of Adverse Environmental Impacts

Construction Phase

Waste Management Hierarchy

6.7.1 The waste management hierarchy has been applied in the assessment and development of mitigation measures for waste which aims at evaluating the desirability of waste management methods and includes the following in descending order of preference:-

- Avoidance and reduction of waste generation;
- Reuse of materials as far as practicable;
- Recovery and recycling of residual materials where possible; and
- Treatment and disposal according to relevant laws, guidelines and good practices.

6.7.2 Based on the waste management hierarchy, waste reduction measures are recommended as follows to reduce impacts and costs arisen from the Project. Recommendations of good site practices and waste reduction measures would be stated in order to achieve avoidance and minimisation of waste generation in the hierarchy. EMP and trip-ticket system are recommended for monitoring management of waste.

Good Site Practices

6.7.3 It is anticipated that adverse impacts would not arise on the construction site, provided that good site practices are strictly followed. Recommendations for good site practices during the construction phase include:-

- Nomination of approved personnel, such as a site manager, to be responsible for good site practices, and making arrangements for collection of all wastes generated at the site and effective disposal to an appropriate facility;
- Training of site personnel in proper waste management and chemical waste handling procedures;
- Provision of sufficient waste reception/ disposal points, of a suitable vermin-proof design that minimises windblown litter;

- Arrangement for regular collection of waste for transport off-site and final disposal;
- Appropriate measures to minimise windblown litter and dust during transportation of waste by either covering trucks or by transporting wastes in enclosed containers;
- Regular cleaning and maintenance programme for drainage systems, sumps and oil interceptors;
- A recording system for the amount of wastes generated, recycled and disposed (including the disposal sites) should be proposed; and
- Preparation of a WMP as a part of the EMP in accordance with ETWB TCW No. 19/2005 Environmental Management on Construction Sites and submit to the Engineer / Architect for approval.

6.7.4 In order to avoid illegal dumping/fly tipping and landfilling, monitoring the delivery of C&D material at landfills and public fill reception facilities, as appropriate, a trip-ticket system should be included as one of the contractual requirements to be implemented by the Environmental Team undertaking the monitoring and auditing work. Warning signs should also be displayed to remind the designated disposal sites. CCTV should be installed at the vehicular entrance and exit of the site as additional measures to prevent illegal dumping/fly-tipping. One may refer to DEVB TCW No.6/2010 Trip Ticket System for Disposal of Construction & Demolition Materials for details. Additionally, all dump trucks should be equipped with GPS or equivalent system for the monitoring of their travel routings and parking locations to prohibit illegal dumping and landfilling of C&D materials. A recording system for the amount of waste generated, recycled and delivery (including disposal sites) as well as travel routings and parking locations should be proposed.

Waste Reduction Measures

6.7.5 Good management and control of construction site activities / processes can minimise the generation of waste. Waste reduction is best achieved at the planning and design stage, as well as by ensuring the implementation of good site practices. Recommendations to achieve waste reduction include:-

- Segregate and store different types of construction related waste in different containers, skips or stockpiles to enhance reuse or recycling of materials and their proper disposal;
- Provide separate labelled bins to segregate recyclable waste such as aluminium cans from other general refuse generated by the work force, and to encourage collection by individual collectors;
- Recycle any unused chemicals or those with remaining functional capacity;
- Maximise the use of reusable steel formwork to reduce the amount of C&D material;
- Wood, steel and other metals shall be separated for re-use and / or recycling prior to disposal of C&D waste to minimise the quantity of waste to be disposed of to landfill;

- Adopt proper storage and site practices to minimise the potential for damage to, or contamination of, construction materials;
- Plan the delivery and stock of construction materials carefully to minimise the amount of surplus waste generated;
- Adopt pre-cast construction method instead of cast-in-situ method for construction of concrete structures as much as possible; and
- Minimise over ordering of concrete, mortars and cement grout by doing careful check before ordering.

6.7.6 In addition to the above measures, other specific mitigation measures are recommended below to minimise environmental impacts during handling, transportation and disposal of wastes.

Storage, Collection and Transportation of Waste

6.7.7 Storage of materials on site may induce adverse environmental impacts if not properly managed. The following recommendations should be implemented to minimise the impacts:-

- Waste, such as soil, should be handled and stored well to ensure secure containment, thus minimising the potential of pollution;
- Storage areas should be maintained and cleaned routinely;
- Stockpiling area should be provided with covers and water spraying system to prevent materials from wind-blown or being washed away; and
- Different locations should be designated to stockpile each material type to avoid cross-contamination and enhance reuse.

6.7.8 Waste collectors should be employed for the collection and transportation of waste generated. For chemical waste, licensed chemical waste collectors for transportation and disposal should be employed in accordance with the Waste Disposal (Chemical Waste) (General) Regulation. The following recommendations should be implemented to minimise the impacts:

- Remove waste in a timely manner;
- Employ trucks with cover or enclosed containers for waste transportation;
- Obtain relevant waste disposal permits from the appropriate authorities;
- Dispose of waste at licensed waste disposal facilities; and
- Maintain records of quantities of waste generated, recycled and disposed.

Construction and Demolition Material

6.7.9 The generated C&D materials should be sorted on-site into inert C&D materials and non-inert C&D waste. To minimise the impact resulting from collection and transportation of C&D materials as far as practicable, C&D waste, such as wood, plastic, steel and other metals should be reused or recycled and, as a last resort, disposed to landfill. Inert C&D materials will be recycled and reused as far as practicable. With

reference to **Table 6.7**, inert C&D materials are expected to be delivered to Tuen Mun Area 38 Fill Bank. A C&DMMP would be prepared in accordance with Section 4.1.3 of the PAH and submitted to PFC for approval in the detail design stage. Within the stockpiling areas, the following measures should be taken to control potential environmental impacts or nuisance:

- Proper handling and storage of waste by means of covers and/or water spraying system to minimise the potential environmental impact and to prevent materials from wind-blown or being washed away;
- Covering materials during heavy rainfall;
- Locating stockpiles to minimise potential visual impacts;
- Warning signs should be displayed at the designated disposal sites. CCTV should be installed at the vehicular entrance and exit of the site as additional measures to prevent fly-tipping;
- Adopting GPS or equivalent system for tracking and monitoring of all dump trucks for the Project in recording their travel routings and parking locations to prohibit illegal dumping and landfilling of C&D materials; and
- Minimising land intake of stockpile areas as far as possible.

Yard Waste

6.7.10 For yard waste, this should be handled in accordance with the principles of reduce, reuse, and recycle (3Rs). Specifically, to minimise the generation of yard waste, the project proponent shall:

- Avoid unnecessary removal or excessive pruning of trees. Preserve trees in their original locations and implement tree transplanting when on-site preservation is not feasible.
- Segregate various types of yard waste and shred wood into smaller pieces if necessary to facilitate reuse and recycling.
- Reuse yard waste (e.g. stumps, pure twigs, leaves and grass clipping, etc.) on-site for a variety of purposes (e.g. decomposition and composting, recreational and decorative uses, and mulching in planting areas, etc.).
- Identify recycling options (e.g. delivery to Y·Park) for yard waste that cannot be directly reused on-site. Minimising land intake of stockpile areas as far as possible.

6.7.11 Where yard waste generation is unavoidable, sorting of yard waste for recycling and reuse on-site should always be prioritised. Yard waste shall be separated from C&D materials to facilitate recycling such as delivering them to Y·Park so as to minimise the quantity of waste to be disposed of at landfill site. Where appropriate, the Contractor should be responsible to cut and shred the yard waste in order to meet the collection requirement of the recycling outlet for processing. Disposal of yard waste directly at landfills should only be regarded as the last resort when no alternatives are available.

6.7.12 In addition, the architect / engineer should prioritise materials with higher recycled content or those that are more easily recyclable, such as engineered wood products over solid lumber. Procuring materials in standardised sizes can also help reduce the

need for on-site cutting and trimming, which often generates significant wood waste. Establishing take-back agreements with suppliers for unused or damaged goods is another effective way to prevent these materials from ending up in landfills. Dedicated storage areas for yard waste in construction sites enables efficient sorting and facilitates recycling and reuse. Ultimately, reducing yard waste in construction requires a holistic approach that spans material selection, procurement, on-site management, and end-of-life processing.

Excavated Sediments

6.7.13 Excavated sediment should handled, treated, transported and/or disposed of in a manner that would minimise adverse environmental impacts. Beneficial reuse will be considered before the disposal of excavated sediment. In order to minimise off-site disposal, the excavated sediment is proposed to be reused on site under the Project (e.g. as backfilling materials). The excavated sediment would be treated by cement stabilisation / solidification (S/S) to improve the material strength to render it suitable for reuse. Treated materials will be subject to Unconfined Compressive Strength (UCS) and Toxicity Characteristic Leaching Procedure (TCLP) testing with a UCS acceptance criteria of ≥ 1000kPa and TCLP meeting the Universal Treatment Standards (UTS) in Title 40 of the Code of Federal Regulations (CFR). If treated materials do not meet the UCS and TCLP acceptance criteria, the materials will be re-treated and re-tested. Pilot-scale trials should be carried out before the commencement of treatment to determine the cement/sediment ratio and curing time to achieve the necessary strength. Off-site sediment disposal should only be considered as the last resort in case sediment treatment / reuse not possible and should follow the procedures in the PAH.

6.7.14 Requirements of the Air Pollution Control (Construction Dust) Regulation, where relevant, shall be adhered to during excavation, treatment, transportation and/or disposal of the sediment.

6.7.15 Excavated sediment treatment areas for carrying out the cement S/S process and any temporary stockpiling should be confined. Runoff should be collected and discharged according to the Water Pollution Control Ordinance (WPCO).

6.7.16 To minimise the potential odour / dust emissions during excavation and transportation of the sediment, the excavated sediments shall be wetted during excavation / material handling and shall be properly covered when placed on trucks or barges. Loading of excavated sediments to barges (if used) shall be controlled to avoid splashing and overflowing of the sediment slurry to the surrounding water.

6.7.17 Should marine disposal of sediment by unavoidable at later stages upon exhaustion of reuse options, separate submissions (e.g. Sediment Sampling and Testing Plan (SSTP) and Sediment Quality Report (SQR)) shall be submitted to EPD's Marine Dumping Control Section / Territorial Control Office if application of a dumping permit under DASO is required. The rationale for sediment removal/disposal will also need to be submitted to the Marine Fill Committee (MFC)/CEDD for agreement in accordance with PAH for Civil Engineering Works, Section 4.2.1.

6.7.18 Stockpiling of sediment requiring Type 2 – Confined Marine Disposal or Type 3 – Special Treatment / Disposal (if any) should be avoided as far as possible. If temporary stockpiling of these sediments is necessary, the excavated sediment should be covered by tarpaulin and the area should be placed within earth bunds or sand bags to prevent leachate from entering the ground, nearby drains and surrounding water bodies. The stockpiling areas should be completely paved or lined in order to avoid contamination to underlying soil or groundwater. Separate and clearly defined areas should be

provided for stockpiling of contaminated and uncontaminated materials. Leachate, if any, should be collected and discharged according to the WPCO.

6.7.19 Barge transporting sediments to designated disposal sites should be equipped with tight fitting seals to prevent leakage and should not be filled to a level that would cause overflow of materials during loading or transportation. In addition, monitoring of the barge loading shall be conducted to ensure that loss of material does not take place during transportation. To avoid illegal dumping, all dumping vessels have to be approved in a marine dumping permit issued under the DASO. Each of the vessels has to be installed with an automatic recording equipment, namely the Front End Mobile Unit (FEMU), which is a key component of the Real Time Tracking & Monitoring of Vessel (RTTMV) System of EPD. The FEMU transmits self-monitoring data direct from the barge at sea to the Control Centre at EPD through GPRS mobile communication network. The transportation route avoiding the ecological sensitive areas should be proposed when applying the dumping permit.

Chemical Waste

6.7.20 If chemical waste is produced at the construction site, the Contractor will be required to register with the EPD as a chemical waste producer and to follow the guidelines stated in the *Code of Practice on the Packaging, Labelling and Storage of Chemical Wastes*. Suitable containers compatible with the chemical waste should be used, and incompatible chemicals should be stored separately. Appropriate labels should be securely attached on each chemical waste container indicating the corresponding chemical characteristics of the chemical waste, such as explosive, flammable, oxidising, irritant, toxic, harmful, corrosive, etc. The Contractor shall employ a licensed chemical waste collector to transport and dispose of the chemical wastes to the licensed CWTC, or other licensed facilities, in accordance with the *Waste Disposal (Chemical Waste) (General) Regulation*.

General Refuse

6.7.21 General refuse should be stored in enclosed bins or compaction units separated from C&D materials. A reputable waste collector should be employed by the contractor to remove general refuse from the site on daily basis. An enclosed and covered area is preferred to reduce the occurrence of wind-blown light materials.

6.7.22 The recyclable component of general refuse, such as food waste, aluminium cans, paper and cleansed plastic containers shall be separated from other wastes. Provision and collection of recycling bins for different types of recyclable waste shall be set up by the Contractor. The Contractor shall also be responsible for arranging recycling companies to collect these materials.

6.7.23 The Contractor shall carry out an education programme for workers in avoiding, reducing, reusing and recycling of materials generation. Posters and leaflets advising on the use of the bins shall also be provided in the sites as reminders.

6.7.24 **Table 6.8** provides a summary of the various waste types likely to be generated during the construction works and operation of the Project, together with the recommended handling and disposal methods.

Table 6.8 Summary of Estimated Waste Arising and Recommended Outlet

Waste Material Type	Generated From	Materials Generated	Total Quantity Generated (Approx.)	Handling Methods	Recommended Outlet
C&D Materials	Site clearance, excavation / foundation works, slope works, piling works, superstructure works etc.	Inert C&D materials – soil, rock, concrete	307,200 m^3	• Segregation from non-inert C&D materials during stockpiling and transportation • Reused on-site as far practicable • Other potential projects / outlets continue to be explored throughout the project to maximise the quantity of the reuse of inert C&D materials	• Suitable material to be sorted and reused on site as much as possible. • The remaining surplus excavated materials would be delivered to Tuen Mun Area 38 Fill Bank for reuse by other project or other construction sites with potential fill demand.
		Non-inert C&D materials – vegetation, timber, papers & plastics	5,900 m^3	• Segregation from inert C&D materials during stockpiling and transportation • Reusable materials should be separated and recycled as far as practicable • Yard waste should be separated and	• Suitable material to be sorted and recycled on site as much as possible. • Yard waste to be reused/recycle before deliver to recycling facility (such as Y·Park) • Materials that cannot be recycled would be

Waste Material Type	Generated From	Materials Generated	Total Quantity Generated (Approx.)	Handling Methods	Recommended Outlet
				reused / recycled as far as practicable	disposed of at the NENT Landfill
Excavated Sediments	Construction of underpass and depressed road	Excavated sediment (included pond sediment, marine deposits and estuarine deposits) from construction of underpass and depressed road at LMC Meander, the Loop and site area between LMC Meander and Ma Tso Lung Road	100,700 m^3	• Treatment by cement S/S technique to treatment standards (UCS ≥ 1000kPa & TCLP ≤ UTS) and reuse on site	On site reuse (e.g. as backfilling materials)
Chemical Waste	From plant, equipment and machinery maintenance and servicing	Cleaning fluids, solvents and lubrication oil	A few hundred litres per month	• Store in compatible containers in designated area on site • To be collected and/or recycled by a licensed chemical waste collector	Recycled by licensed facility and/or disposal of at the CWTC
General Refuse	Resident site staff and workers	Food waste, plastic bottles, aluminium cans and wastepaper	Approximately 65 kg/day	• Reuse or recycle as much as possible before disposal of at landfill • Provide on-site collection points	Approved waste transfer or disposal facilities and then to WENT / NENT Landfill

Waste Material Type	Generated From	Materials Generated	Total Quantity Generated (Approx.)	Handling Methods	Recommended Outlet
				together with recycling bins • To be collected by an appropriate waste collector to be employed by the Contractor	

Operation Phase

Chemical Waste

6.7.25 Requirements in the Code of Practice on the Packaging, Labelling and Storage of Chemical Wastes would be followed in handling of chemical waste as in the construction phase. A trip-ticket system would be adopted by the operator to monitor disposal of chemical waste. The Contractor shall register with EPD as a chemical waste producer. The licensed chemical waste collector shall deliver the waste to the CWTC at Tsing Yi, or other licensed facility, in accordance with the Waste Disposal (Chemical Waste) (General) Regulation.

General Refuse

6.7.26 Recycling of wastepaper, aluminium cans and plastic bottles should be encouraged, it is recommended to place clearly labelled recycling bins at designated locations which could be accessed conveniently. Other general refuse should be separated from chemical waste by providing separated bins for storage to maximise the recyclable volume.

6.7.27 The amount of general refuse will be insignificant and will be removed during maintenance works.

6.8 Evaluation of Residual Impacts

6.8.1 With the implementation of the recommended mitigation measures for the handling, transportation and disposal of the identified waste types arising, no adverse residual impact would be anticipated.

6.9 Environmental Monitoring and Audit Requirements

6.9.1 It would be the contractor's responsibility to ensure that any wastes produced during the construction and demolition works are handled, stored and disposed of in accordance with good waste management practices and relevant regulations and other legislative requirements. The recommended mitigation measures should form the basis of the site WMP to be developed by the Contractor in the construction stage.

6.9.2 It is recommended that the waste arisings generated during the construction activities should be audited regularly by the Environmental Team (ET) to determine if they are managed in accordance with approved procedures. The audits should look at all aspects of on-site waste management practices including waste generation, storage, recycling, transport and disposal. Apart from site inspections, documents including licenses, permits, disposal and recycling records should be reviewed and audited for compliance with the legislation of the recommended good site practice and other waste management mitigation measures.

6.9.3 A WMP, as a part of the EMP, should be prepared in accordance with ETWB TC (W) No.19/2005 and submitted to the Engineer for approval. The recommended mitigation measures should form the basis of the WMP. The monitoring and auditing requirement stated in ETWB TC (W) No.19/2005 should be followed with regard to the management of C&D materials.

6.9.4 No EM&A requirement is considered necessary during the operation phase.

6.10 Conclusion

6.10.1 The main waste types generated during the construction of the Project would be C&D materials (from site clearance, excavation/foundation works, slope works, etc.), excavated sediment, chemical waste from plant operation and maintenance and general refuse from workforce. Provided that these wastes are handled, reused, transported and disposed of using approved methods and that the recommended good site practices are strictly followed, no adverse environmental impacts are anticipated.

6.10.2 During the construction of the Project, the quantity of C&D materials generated during the construction phase is estimated to be 313,100 m^3 with 5,900 m^3 of non-inert materials and 307,200 m^3 of inert materials. It is estimated that of the inert C&D materials generated, approximately 15,700 m^3 would be reused on site as backfill materials. Surplus inert C&D material will be exported off-site to other construction sites or delivered to PFRF (Tuen Mun Area 38 Fill Bank). A minimal amount (in the order of a few hundred litres per month) of chemical waste and 65 kg of general refuse are estimated to be generated daily. Chemical waste would be recycled at an appropriate facility as far as possible, while chemical waste that cannot be recycled should be disposed of at CWTC, or another licensed facility. General refuse would be disposed of at WENT / NENT Landfill.

6.10.3 It is estimated that 100,700 m^3 of excavated sediment will be generated from the Project during underpass / depressed road excavation. Excavated sediment will be treated by cement S/S to improve its strength characteristics and following confirmatory UCS and TCLP testing, will be reused on site (e.g. as backfilling materials).

6.10.4 With the implementation of the recommended mitigation measures detailed in **Section 6.7**, no adverse environmental impacts would be expected from excavation and handling of sediments.

6.10.5 During operation of the Project, A minimal amount of chemical waste and general refuse are estimated to be generated during each maintenance event.

6.10.6 No adverse environmental impact (including potential hazard, air and odour emissions, noise, wastewater discharge and public transport) is anticipated by the handling, collection, transportation and disposal of the different types of waste.